Mapping "Cult," Second Pass: The v2 Geometric Analysis

Working report – draft base material for the Results chapter, with a first attempt at interpretation

Prepared for Célestin Meunier's thesis project

Covers run `20260910-101829`, generated 2026-09-10

How to read this document. This is still not a finished Results chapter. Sections 2–11 report numbers, not arguments. Section 12 is a first, explicitly provisional pass at what these patterns *might* mean, kept structurally separate and visually marked (a tan box) throughout so a reader can tell at a glance which sentences are geometric fact and which are a reading of that fact the researcher is free to revise or reject. Green boxes mark a specific, checkable methodological correction made during this run, with what was wrong and how it was verified fixed – included because the correction itself is informative about how much to trust a "count of named things" figure in general. Nowhere in this document is geometric proximity treated as evidence that any group, practice, or person *is* or *is not* "a cult."

Contents

1 In short: what did this run find?

- **The corpus got much smaller, on purpose** (same as the first v2 pass): a stricter, judge-verified extraction pipeline cut literature from 39,236 to 5,741 expressions, trading quantity for verified fidelity.
- **The three sources still barely form separate geometric clusters** – silhouette near zero, same conclusion as before.
- **A real bug was found and fixed in how "emergent entities" (named groups/leaders the corpora talk about) were counted.** The first v2 pass found only 21 entities, almost all mentioned exactly once, zero shared across all three corpora. That was not a fact about the corpus – it was because entity extraction only looked inside expressions that survived the strict extraction filter, silently discarding named-entity mentions sitting in text that didn't. Fixed, this run finds **753** entities, including **3 mentioned by all three corpora at once** ("New Age," Jehovah's Witnesses in French, and the Order of the Solar Temple) and **55 mentioned only in interviews** (up from 0). Full story in Section 6.
- **Named things still sit at the edge of their own corpus's "typical" region; generic statements sit near the centre** – same pattern as before, unaffected by the entity fix (it concerns ordinary expressions, not the entity layer).
- **"Sept?" has been manually removed from the data** at the researcher's explicit instruction (Section 5) – not just flagged as before. Excluded at pooling time, never by

editing the raw extraction archive.

- **New this run: where is the entity layer dense, where is it empty, and does that matter?** HDBSCAN clustering (the real, 393-dimensional space, never a 2-D projection) finds **49** density-based clusters among the 753 entities, with **514** left as noise (not unimportant – just not sitting inside a dense neighbourhood at this threshold). A density/gap map over a 2-D projection flags **9 of 167** in-bounds grid cells as candidate coverage gaps. Full story in Section 7.
- **Three ways to partition that space (Voronoi), tried and compared, not just one chosen in advance.** Only one of the three worked as a legible 2-D picture – and the reason why is itself an informative, honestly-reported finding, not hidden. Section 8.
- **New this run: literature's publication year barely correlates with where a document's expressions sit** (Spearman $\rho = -0.064$, statistically significant at this sample size but a very weak effect) – **and the single nearest MIVILUDES criterion to literature's "centre of gravity" is the same one in every one of the three publication-year periods checked.** Section 10.

2 Where the project stands

2.1 Why a second pass at all

A 100-item manual fidelity review of the first extraction pipeline (2026-09-08) found it systematically over-inclusive: paraphrased or mistranslated expressions, document headings and bibliography citations extracted as though they were claims, occasional interviewer questions mislabelled as participant statements. The pipeline (**v2**) was rebuilt around a strict verbatim-span rule plus a second, independently- prompted model that judges every retained candidate the way a human reviewer would ("model-judged," never a substitute for human review). v1's own archives and every retained run stay frozen on disk, untouched.

2.2 Stage 1–2 – Same three corpora, far fewer expressions

Same underlying documents as v1 (57 scholarly works, 2 MIVILUDES documents, 26 interview transcripts); far stricter extraction:

Corpus	v1 expressions	v2 expressions	Change
literature	39,236	5,741	−85.4%
MIVILUDES	914	108	−88.2%
interviews	230	64	−72.2%

2.3 Stage 3–4 – Reference vocabularies unchanged; emergent entities, twice

The three reference vocabularies (concept backbone, structural concepts, ConceptNet-generalised concepts) are unchanged from v1, reused as-is. Emergent entities were built *twice* this session – the first attempt undercounted badly; see Section 6 for the full story and the fix.

2.4 Stage 5 – One shared geometric space, rebuilt

Same pooling procedure as v1: embed everything with the same model, standardize, PCA down to the smallest dimensionality keeping 95% of the variance. With the corrected, much larger emergent-entities layer, this run's space holds **11,377** points, compressed to **393** dimensions

(95.0% variance kept) – up from the first v2 pass's 10,646 points/389 dimensions, mostly due to the extra 732 entity points now correctly included, minus the single "Sept?" point manually excluded this run (Section 5).

2.5 Stage 6 – The interview "first impression" layer: ported, not redone

Unchanged from the first v2 pass: the 25 manually-verified interview opening-prompt exemplars are re-projected through this run's transform rather than re-selected – the embedding of a quote is a property of its own words, not of which build happened to be running.

2.6 Stage 7 – The equal-n correction still applies, at a smaller scale

Unchanged finding from the first v2 pass: literature is still **97.09%** of expression points (the emergent-entities fix does not touch expression counts), so the same equal-n/equal-size corrections are still necessary and still used, drawing 64 points per source (down from v1's 204).

2.7 Stage 8 – Two toolkit runs, one throwaway

The reusable 10-module analysis toolkit was run against this shared space twice this session: once against the (bugged) 21-entity space, and again – covered by this document – after the fix, against the corrected 753-entity space (run `20260910-010254`). The first run's own report is not being kept as a separate document; this report supersedes it. A fresh spot-check sample (Section 5) was drawn from the retained v2 extraction output before either toolkit run and is unaffected by the entity-layer fix (it concerns expression extraction, not entities).

3 Key ideas, explained in plain language

Unchanged from earlier documents – included again so this one is self-contained. Skip if already familiar.

3.1 Embeddings, distance, and similarity

An **embedding** converts a piece of text into a long list of numbers (a **vector**, 1,024 numbers here) such that similar meanings end up as nearby vectors. **Euclidean distance** is ordinary straight-line distance in that many-dimensional space (smaller = more similar). **Cosine similarity** instead measures whether two vectors point in the same *direction* from the origin, ignoring length (-1 opposite, 1 identical direction, 0 unrelated). Both are always reported side by side.

3.2 Centroid, standardization, and PCA

A **centroid** is a group of points' average position – literature's "average position," MIVILUDES's, and so on. Every dimension was **standardized** before pooling so no source dominates purely because its raw numbers are larger. **PCA** then compresses the 1,024-number description of each point down to the smallest number of dimensions keeping 95% of the original variation – 393 dimensions this run. Every distance/similarity number below is computed in this 393-dimensional space.

3.3 UMAP, k-NN composition, and silhouette

UMAP produces an approximate, readable 2-D layout purely for visualisation – not itself evidence. A point's **k nearest neighbours** are the k closest other points (k=10, 20 here);

averaging their source-composition over many points asks how much different sources overlap versus occupy separate regions. **Silhouette** scores (roughly -1 to 1) how well a proposed grouping matches the data's actual geometric clustering; near 0 means the groups don't correspond to distinct clusters.

3.4 Bootstrap confidence intervals; equal-size/equal-n comparisons

"Mean, 95% CI [low, high]" comes from **bootstrap resampling**: repeating a measurement 100 times on randomly resampled subsets. The three expression sources (5,741/108/64 this run) and the three reference vocabularies (3,000/1,500/195) are wildly different sizes; an **equal-size-controlled** comparison down-samples every set to the smallest before comparing, controlling for pool-size bias specifically – not evidence the sets are otherwise interchangeable.

3.5 Typicality and "borderline" points

A point's **typicality** is simply how close it sits to its own group's centroid – close = "typical," far = "atypical," a purely geometric fact about this space under this model, not a claim about cognitive prototypicality or real-world category membership. A **borderline** point between groups A/B is one whose distances to each centroid are nearly equal.

3.6 "Point role," "source dataset," and provenance categories

source_dataset: which of the eight underlying datasets a point came from. **point_role**: expression / reference / emergent. Every emergent entity also carries a **provenance category** – `literature_only`, `miviludes_only`, `interviews_only`, `shared_2_corpora`, or `shared_all_3` – describing only which corpora mention it, not any normative claim about the entity itself.

4 The datasets pooled into the shared space

Source	Points	Role	What it is
literature	5,741	expression	Strict-verbatim extraction, 57 scholarly works
miviludes	108	expression	Strict-verbatim extraction, 2 MIVILUDES documents
interviews	64	expression	Strict-verbatim extraction, 26 interview transcripts
miviludes_criteria	17	expression	MIVILUDES's 17 official sectarian-drift criteria (unchanged)
concept_backbone	3,000	reference	Topic-neutral WordNet-derived concepts (unchanged)
structural_concepts	1,500	reference	Corpus-mined structural/relational vocabulary (unchanged)
conceptnet_concepts	195	reference	Hand-reviewed ConceptNet generalisation (unchanged)
emergent_entities	**753**	emergent	Named entities, corrected this run – see Section 6

Table 1: The eight sources pooled into the 11,377-point, 393-dimensional v2 shared space.

A ninth, separate layer – 25 ported interview prototypes – sits in the same coordinate system but is not part of this count or the PCA fit.

5 Spot-check: is this run's data trustworthy?

Carried over unchanged from the first v2 pass – this concerns expression extraction quality, not the entity-layer fix below, so it does not need repeating. A fresh random sample was drawn from each corpus's final, judge-accepted output (100 literature, 40 MIVILUDES, 30 interview expressions; seed 42; full methodology in `processed/audits/v2_spotcheck_20260909.md`). Automated checks found 0 verbatim/faithfulness failures across 170 sampled rows. An independent read found expressions correctly attributed and sensibly tagged throughout, with one genuine, small finding:

One genuine finding: "Sept?". Interview `b3-aug23-1213` (participant, age 97), asked in French "*Quand je te dis le mot secte à quoi tu penses?*" ("When I say the word 'secte' [cult], what do you think of?"), answered **"Sept?"** ("seven?") – almost certainly a mishearing of *secte* as *sept*, not a substantive response. The judge model scored it faithful/self-contained/cult-relevant/atomic, correctly noting it is a genuine transcript quote – true, but the string carries no propositional content about cults. Scale: 1 of 11,378 points (0.01%).

Update, this run: at the researcher's explicit instruction, this point has been manually removed from the shared space – not merely flagged. The removal is a single, individually-documented exclusion applied at pooling time (`MANUALLY_EXCLUDED_POOLED_KEYS` in `build_shared_space.py`, keyed `"b3-aug23-1213:0"`), never by editing the raw extraction archive – the archive itself stays a complete, honest record of what the pipeline actually produced, imperfections included; only the pooled analysis space excludes this one point. The space now holds 11,377 points, not 11,378. No other result in this document changed detectably from this single point's removal.

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6 Emergent entities: found, broken, and fixed

This section gets its own, expanded treatment because the correction made here is the main methodological story of this run, and because understanding *why* it was wrong is useful for judging how much to trust the corrected number.

6.1 What went wrong

The first v2 toolkit run found only **21** emergent entities – 20 literature-only, 1 miviludes-only, 0 interviews-only, 0 shared by any two or three corpora – almost all mentioned exactly once. This looked implausible on inspection: several of the 21 were not really named entities at all ("leadership," "desert," "post-Soviet remaking of identities" – generic phrases, not named things), and the corpus was known independently (Section 5, the miviludes/interviews domain_terms counts below) to discuss dozens of well-known groups by name.

Root cause. Entities were being extracted only from `entity_anchors`, a field the extraction model fills in *only for expressions that already survived* the strict verbatim+judge screening (`extraction_v2_schema.py`: "named groups, leaders, movements mentioned **in the span**"). Since that screening now keeps only 14–28% of what v1 kept per corpus, roughly 75–85% of named-entity mentions sitting inside now-rejected spans were silently discarded – even when the surrounding text discussing that entity was perfectly legitimate, just not judged a standalone *claim* worth keeping.

The fix uses a second field the extraction model already produces but that was never wired into the shared space: `domain_terms` – "cult-related concepts or named groups that appear in the text, even if no expression around them is worth keeping." Checked directly: literature alone has 52,179 unique `domain_terms` (100,031 mentions) against the 20 `entity_anchors`-only entities previously used.

6.2 Getting to a clean list: two more problems, found and fixed

`domain_terms` is a much broader net than `entity_anchors` – it mixes genuine named entities ("Scientology," "Heaven's Gate") with generic domain vocabulary ("cults," "brainwashing," "mind control") that already has its own home in the structural_concepts/concept_backbone reference vocabularies. Two further checks against real data were needed before the result was trustworthy:

Casing filter, and a bug in it. Checked directly: `domain_terms` terms in their original, un-lowercased casing split cleanly – capitalized terms are almost entirely genuine named entities (ISKCON, Branch Davidians, UNADFI); all-lowercase terms are almost entirely generic vocabulary (secularization, deprogramming, mind control). A simple "not all lowercase" filter was applied to *both* `domain_terms` and (newly) `entity_anchors` – the latter had never had any type filtering at all, which is exactly why v1's top-mentioned "entity" was "charismatic leader" (3,417 mentions, 2nd highest overall) and why v2's first pass still let "leadership"/"desert" through. One bug found in the filter itself: Python's `str.islower()` is `False` for a string with *no* cased characters at all, so bare years and page numbers ("2004," "11") were passing as if capitalized. Fixed by requiring at least one actual letter.

Cited-author contamination. Even after the casing fix, the single largest remaining source of noise in literature was bare surnames of *this corpus's own cited authors* – academic citations like "(Richardson 1985)" leave a bare capitalized surname in the text, indistinguishable by casing or frequency alone from a genuine named entity. Checked directly: "richardson" alone had 47 mentions (more than most real cult groups); "barker" 34; "anthony" 28. A frequency threshold cannot separate self-citation from a genuinely important recurring entity – both look identical (a capitalized word mentioned often). Fixed with a targeted, data-derived stop-list: every author surname is read directly from `metadata/literature.csv` (the corpus's own 57-document bibliography, 90 unique surnames) and excluded – no hardcoded name list, automatically correct if the corpus grows. **Not** fixed: externally-cited classical theorists who are not themselves corpus authors ("weber," "freud," "wallis" all still pass) – flagged as a known residual limitation, not silently hidden.

6.3 Per-corpus, not global, mention thresholds

A single mention-count threshold applied to the cross-corpus total would still favour literature's citation-heavy volume unfairly: a name mentioned a handful of times there is plausibly just a citation, while the same count in MIVILUDES/interviews' much smaller, curated text is comparatively far more likely to be deliberate. This run uses **per-corpus** thresholds – literature ≥ 10, MIVILUDES ≥ 3, interviews ≥ 1 – and an entity qualifies if it clears *any one* corpus's own threshold.

6.4 The result

Top entities by total mentions are now entirely genuine named groups/movements: NRMs (685), Scientology (299), New Age (270), Heaven's Gate (268), Unification Church (250), Jehovah's Witnesses (211), Branch Davidians (159), Jonestown (156), Solar Temple (136), ISKCON (123), Children of God (118), Theosophy (109), Aum Shinrikyo (103), Satanism

Provenance category	First v2 pass (buggy)	This run (fixed)
literature_only	20	603
miviludes_only	1	26
interviews_only	0	55
shared_2_corpora	0	66
shared_all_3	0	3
Total	**21**	**753**

Table 2: Emergent-entity counts, before and after the fix.

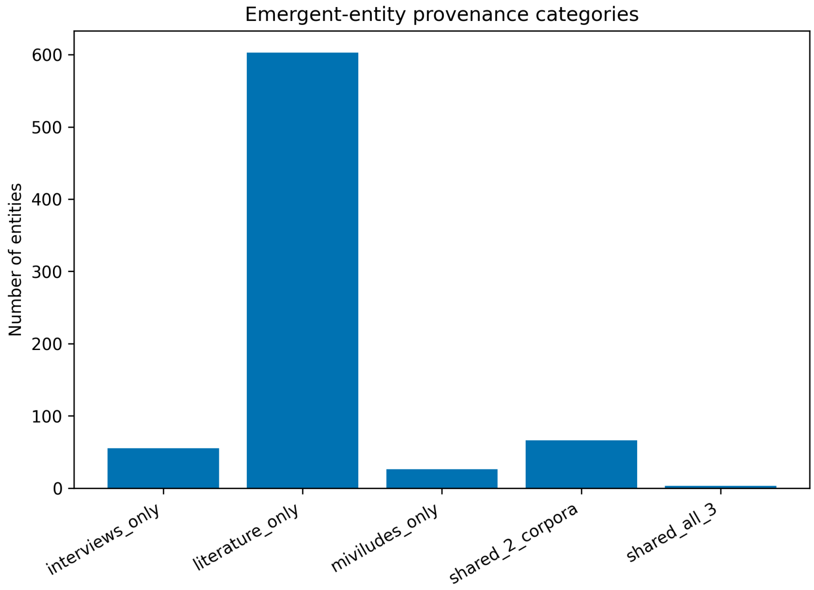

Figure 1: Emergent entities grouped by which corpus/corpora mention them, this run.

(95), Soka Gakkai (86) – contrast v1's top entities, which included "charismatic leader" (2nd highest) and bare generic nouns ("cults," "religion," "church," "followers").

The 3 entities mentioned by all three corpora: "new age" (270 total: 259 literature, 9 miviludes, 2 interviews), "témoins de jéhovah" (French for Jehovah's Witnesses; 9 total: 1/7/1), and "ordre du temple solaire" (French for Order of the Solar Temple; 4 total: 2/1/1). Their nearest-neighbour expressions per corpus (full retrieval, not equal-n sampled) read sensibly: literature's nearest neighbours to "new age" are literally "New Age" mentions; to "témoins de jéhovah," "Jehovah's Witnesses" mentions – a direct cross-language match, not a coincidence of the embedding space. Full tables in `analyze_emergent_entities/shared_anchor_nearest_expressi`

55 entities are now interviews-only (up from 0): Netflix, "AI cult," Blue Öyster Cult, Raëlism, "Tomato cult," "Wild Wild Country," Richard Branson, and others – genuine, previously entirely invisible signal about what interviewees personally associate with the topic, distinct from what the literature or MIVILUDES discuss.

This is not a solved problem. Manual spot-checking the top-N table above and the shared-anchor retrieval found no remaining obvious false positives, but the underlying filters (casing + frequency + a corpus-author stop-list) are not a general citation detector, and the residual externally-cited-theorist gap noted above is real. Treat this as a substantially improved, still-not-fully-vetted candidate list – the same posture as every other automated extraction step in this project – not a hand-verified named-entity list.

As before, this module does **not** classify any entity as "genuinely a cult" versus "mainstream" or apply any other normative label – it reports who talks about what, not who is right.

7 Entity topology: clusters, density, gaps, centrality

New this run. Three questions about the 753 emergent entities specifically, all answered in the REAL 393-dimensional space (never a 2-D projection – pictures come after, as a reading aid, not the evidence itself): do they form dense clusters or sit scattered; how locally dense is each entity's own neighbourhood; and how central or peripheral is an entity/cluster relative to the space's overall "centre of gravity" (the same equal-corpus-weighted grand centroid framing used throughout this project, applied here to entities the way `analyze_typicality.py` already applies it to ordinary expressions).

Governing principle. Cluster membership, local density, and centrality are purely geometric facts about this embedding space under this model. A "dense cluster" of entities is not itself evidence those entities are conceptually similar in any deeper sense than "their extracted-context embeddings landed near each other"; an HDBSCAN "noise" label does not mean an entity is unimportant, only that it did not sit inside a density-defined cluster at the chosen threshold.

`sklearn.cluster.HDBSCAN` (density-based – no pre-specified cluster count, and points that don't sit in a dense neighbourhood are left as explicit "noise" rather than force-assigned) finds **49 clusters** among the 753 entities, at a minimum cluster size of 3; **514 entities (68%)** are noise at this threshold.

Read qualitatively (see Section 12 for a more careful reading): the most central clusters are generic, multilingual core vocabulary for the topic itself (sect/secte/sekte, "new religious movement," Satanism-adjacent terms); the most peripheral are specific named groups and, oddly, a cluster of three unrelated California place names – flagged here as a likely artefact (entities co-occurring in discussion of California-based groups, not a meaningful "cult entity" cluster on its own terms) rather than a finding, consistent with this project's practice of not smoothing over a plausible artefact just because it is a small one.

Coverage gaps: binning that same 2-D layout into a 15×15 grid over its own bounding box and counting entities per cell finds **9 of 167** in-hull cells (cells inside the entities' own convex hull, i.e. genuinely surrounded by data rather than just empty space at the edge of the picture) with zero entities – candidate coverage gaps: regions of "entity space" that sit between occupied areas without any named thing of their own.

8 Voronoi regions: three seed strategies, tried and compared

"Try a few different seed strategies and see what happens," per the researcher's own framing, rather than picking one in advance. All three tessellations are fit on the exact same 2-D UMAP layout of the 753 entities (verified byte-identical member coordinates across all three

Cluster	Size	Dist. to grand centroid	Members
Most central (closest to the space's overall "centre of gravity")			
14	6	20.53	dérives sectaires; jugendsekten; les dérives sectaires; sects; sekte; une secte.
31	14	20.92	center for studies on new religions; new age; new age movement; new religious movement(s); ...
22	10	21.90	church of satan; satan; satanic; satanism; satanist(s); the cult of satan; ...
Most peripheral			
46	6	31.56	catholic; catholic church; catholicism; catholics; roman catholic (church)
8	3	31.65	clapham; clapham sect; the clapham sect
19	3	32.23	moonie; moonies; the moonies
35	3	33.26	california; los angeles; san francisco
34	4	33.32	jehovah's witnesses; les témoins de jéhovah; témoins de jéhovah; zeugen jehovas

Table 3: Most central and most peripheral entity clusters, by distance from the cluster's own member-centroid to the space's grand centroid. Full 49-cluster table in `analyze_entity_topology/cluster_summary.csv`.

runs, since `umap-learn` is deterministic given a fixed seed) – only the *seed points* a Voronoi diagram is built from differ:

- **Criteria-seeded**: the 17 MIVILUDES criteria.
- **Corpus-centroid-seeded**: the 3 corpus centroids.
- **Cluster-centroid-seeded**: the 49 HDBSCAN cluster centroids from Section 7.

What actually happened – reported honestly, not smoothed over. Only the **cluster-centroid-seeded** strategy produced a legible 2-D tessellation (Figure 4). The other two collapsed: all 17 criteria, and separately all 3 corpus centroids, land compressed into one small corner of the 2-D layout, far from the bulk of the entity cloud (Figures 5–6). This is a real, understood, and reportable cause: cluster centroids are computed directly from the SAME entity vectors UMAP was fit on (an "in-distribution" query), while the criteria and corpus centroids come from an entirely different, unrelated point population and only enter the 2-D layout afterward, via `reducer.transform()` – a well-documented UMAP limitation (out-of-sample points can be compressed into an unrepresentative region of the fitted manifold), not evidence that the 17 criteria or the 3 corpora are somehow geometrically confused with each other. The underlying HIGH-DIMENSIONAL retrieval this document relies on elsewhere (nearest entities to each criterion, Section 9.5; nearest entities to each corpus centroid) is computed directly in the real 393-D space and is entirely unaffected by this 2-D projection artefact – only this specific 2-D picture is.

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9 What the rest of the toolkit found

The other nine modules are materially unaffected by the entity-layer fix (they concern expressions, not emergent entities) – numbers shift slightly because the shared space grew by 732 points, but conclusions are unchanged from the first v2 pass. Summarised here; see the first

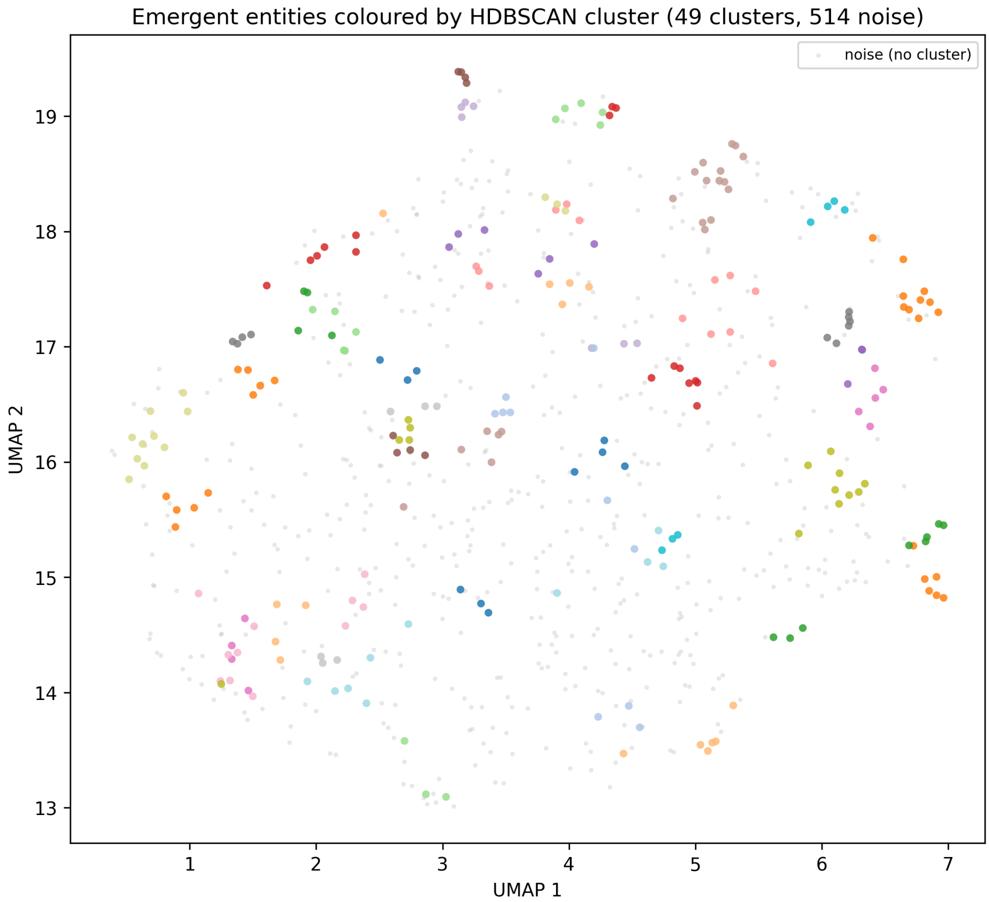

Figure 2: The 753 entities, 2-D UMAP projection, coloured by HDBSCAN cluster (grey = noise). A reading aid only – cluster membership itself comes from the real 393-D space above, not from this picture; shown here because the 2-D layout visually confirms the same compact, well-separated groupings the high-dimensional clustering found, a useful cross-check.

v2 report or the run's own CSVs for full detail.

9.1 Free-listing rank audit

Same verdict: **not usable**. Narrow check ("rank 1 is the participant's own first claim"): 18/25 documents (72.0%). `response_rank` plays no role anywhere in this report.

9.2 Global structure

Selected pairwise centroid distances (full 8×8 matrix in the run's own CSV):

Source A	Source B	Euclidean	Cosine
literature	miviludes	8.00	0.50
literature	interviews	8.14	0.60
miviludes	interviews	11.98	0.22
literature	miviludes_criteria	14.44	0.14
miviludes	miviludes_criteria	12.10	0.51
literature	emergent_entities	11.30	0.53

Essentially unchanged from the first v2 pass's numbers (cosine similarities between the three

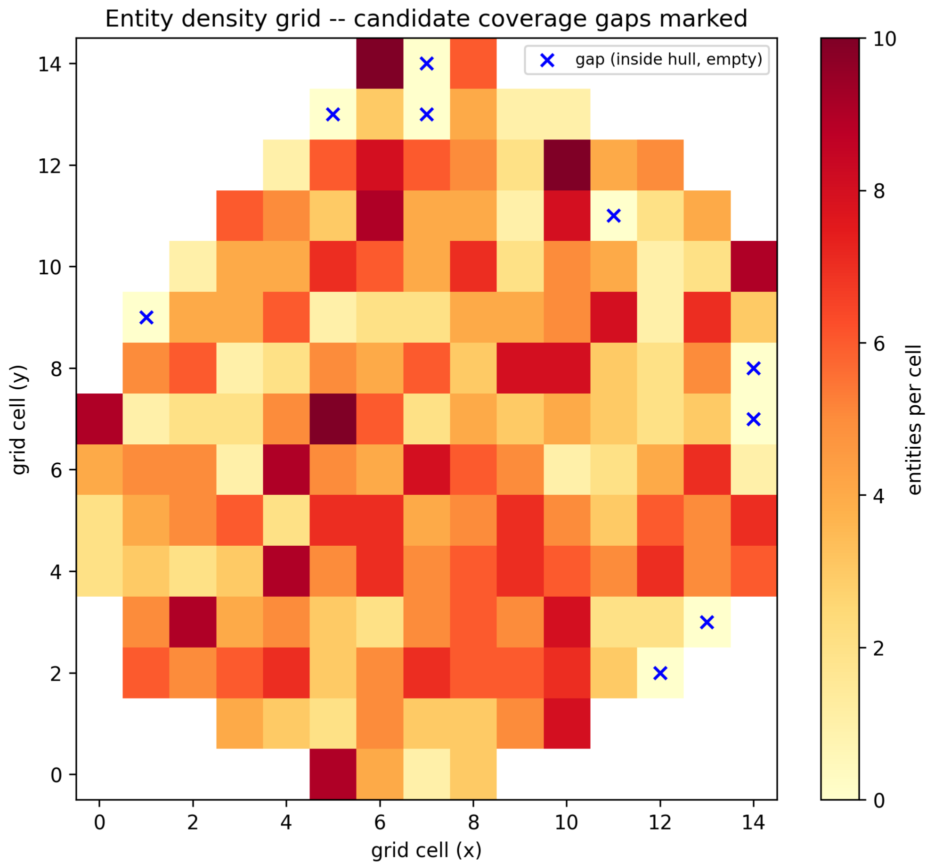

Figure 3: Entity count per grid cell (2-D UMAP layout); blue × marks a candidate gap (inside the entities' convex hull, zero entities).

expression corpora remain solidly positive, 0.22–0.60, versus v1's near-zero/negative values) – consistent with the entity-layer fix affecting the emergent_entities centroid specifically (it moved: distance to the combined-expression centroid is now 11.30, versus the first pass's 10.22, since the entity set itself grew nearly 36-fold and now includes a much wider spread of entities) without materially changing the three expression corpora's own relationship to each other.

9.3 Cluster structure

Silhouette (equal-n, k=10): **0.0130** (95% CI [0.0096, 0.0164]) – indistinguishable from the first v2 pass's 0.0130. The three sources still overlap substantially rather than forming separate clusters.

9.4 Criterion neighbours

Language-representation sensitivity: French-original vs. projected English translation, **3 of 17** criteria change nearest corpus; vs. raw cosine, **4 of 17** – unchanged from the first v2 pass.

9.5 Per-criterion and per-centroid tables: nearest expressions and entities

New this run. `analyze_criterion_neighbours.py`'s existing per-criterion nearest-expression retrieval now has a nearest-*entity* companion (top-5 per criterion, same 393-D space, computed directly – unaffected by Section 8's 2-D projection issue). One example of all 17:

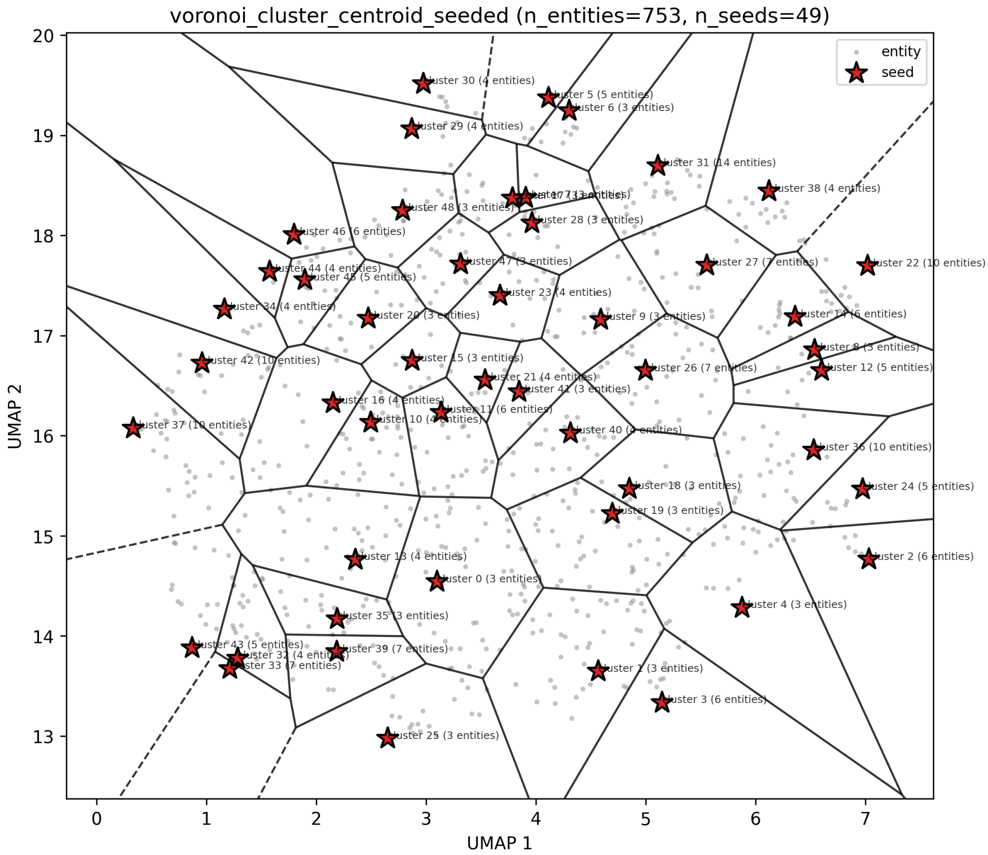

Figure 4: Cluster-centroid-seeded Voronoi tessellation – the one strategy that produced a legible partition, since its seeds are drawn from the same population the 2-D layout was fit on.

Rank	Nearest entity
1	thought reform and the psychology of totalism
2	brainwashing
3	phénomène d'emprise mentale
4	silva mind control
5	spiritism

Table 4: Nearest entities to *crit-mental-destabilization* ("mental destabilization") – one of 17 criteria's own tables, full set in `analyze_criterion_neighbours/qualitative_retrieval_entities.csv`. Each criterion's existing per-criterion focused-projection mini-map (`generate_focused_projections.py`) now overlays these nearest entities alongside the pre-existing nearest expressions and reference terms, one picture per criterion, 17 total, not reproduced individually here for space.

The same treatment was applied to **centroids**, both kinds asked for: the 3 corpus centroids, and the cluster centroids from Section 7.

For the 49 entity clusters, the same nearest-expression/nearest-entity retrieval was run against the 10 largest clusters' own member centroids (by size, a different ordering from Section 7's centrality ranking – kept deliberately separate, since "biggest" and "most central" needn't agree). The largest cluster's own nearest expressions read, unsurprisingly, as direct restatements of its members' own vocabulary ("New Religions," "New Religious Movement(s)," . . .) – full table in `analyze_entity_topology/cluster_centroid_nearest_{expressions,entities}.csv`.

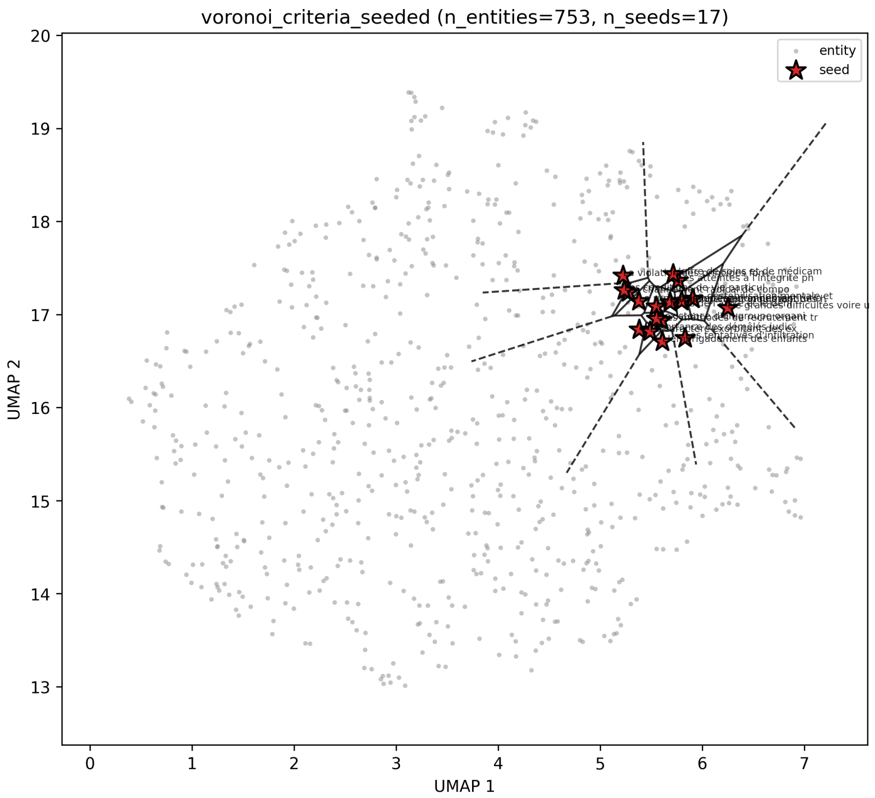

Figure 5: Criteria-seeded attempt – all 17 criteria compress into one corner under UMAP's out-of-sample transform; included for honesty about what was tried, not because it is a useful reading aid on its own.

Corpus centroid	Rank 1 nearest entity	(French items are the corpus's own vocabulary)
literature	une secte.	–
miviludes	une secte.	–
interviews	une secte.	–

Table 5: Nearest entity to each of the 3 corpus centroids – the generic French phrase "une secte." (roughly "a sect.") ranks #1 for *all three* corpora's own centroid, ahead of any specific named group. Full top-10-per-corpus table in `analyze_global_structure/source_centroid_nearest_entities.csv`.

9.6 Interview initial exemplars

Unchanged: same 25 resolved/1 unavailable interviews, same manually-verified quotes, re-measured against this run's space.

9.7 Typicality and borderline points

Unaffected by the entity fix (concerns ordinary expressions). Same pattern as before: the most atypical items per corpus are specific named things (literature: "Jose Silva," "Asahara was born Matsumoto Chizuo in 1955"; miviludes: "Steiner-Waldorf schools," "Scientology"; interviews: "Gourou," "Jehovah's Witnesses"), while the most typical are generic statements ("This is Christianity.," "sectarian deviations," "the cult").

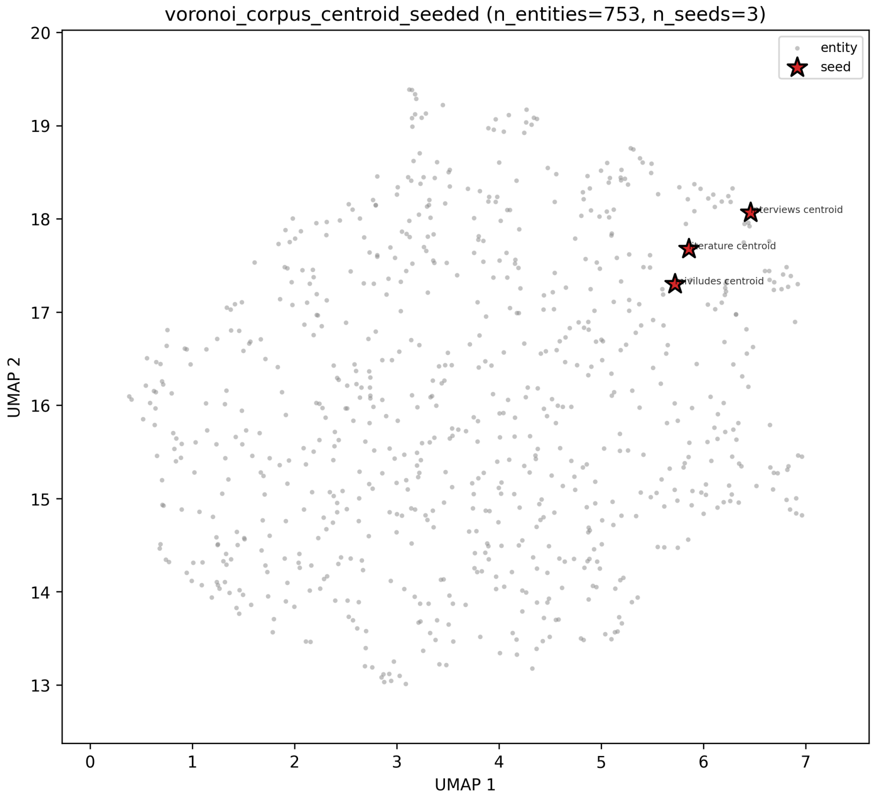

Figure 6: Corpus-centroid-seeded attempt – same compression, worse still with only 3 seeds (scipy needs ≥4 points for a 2-D Voronoi diagram at all, so no cell boundaries are drawn here, seed positions only).

9.8 Focused projections

23 curated populations generated this run, including a new `emergent_entity_focus` population (95 points, up from the first pass's 55 – top 50 entities plus overlay context) now genuinely worth looking at:

10 Publication-date effects (literature only)

New this run. Literature-only: MIVILUDES has no usable per-document year (only a `.bib` file's `date = {2024}` for one of its 2 source documents, nothing structured); interviews have real collection dates but all within one narrow month, not a multi-year span. Year comes directly from each literature `document_id` (e.g. `clarke-2004-encyclopedia-of-new-religious-movements`) – the corpus's own bibliography CSV uses an unrelated Zotero citekey with no join back to the pipeline's document IDs, so this document-id extraction is the only way to recover year at all; verified **5,741/5,741** (100%) literature points have an extractable year this way.

Governing principle. A correlation or period-centroid shift here is a geometric fact about extracted-expression embeddings by publication year; it is not, by itself, evidence of a causal trend in scholarly opinion, a claim about which period's literature is "more correct," or a claim that generalizes beyond this specific 57-document sample.

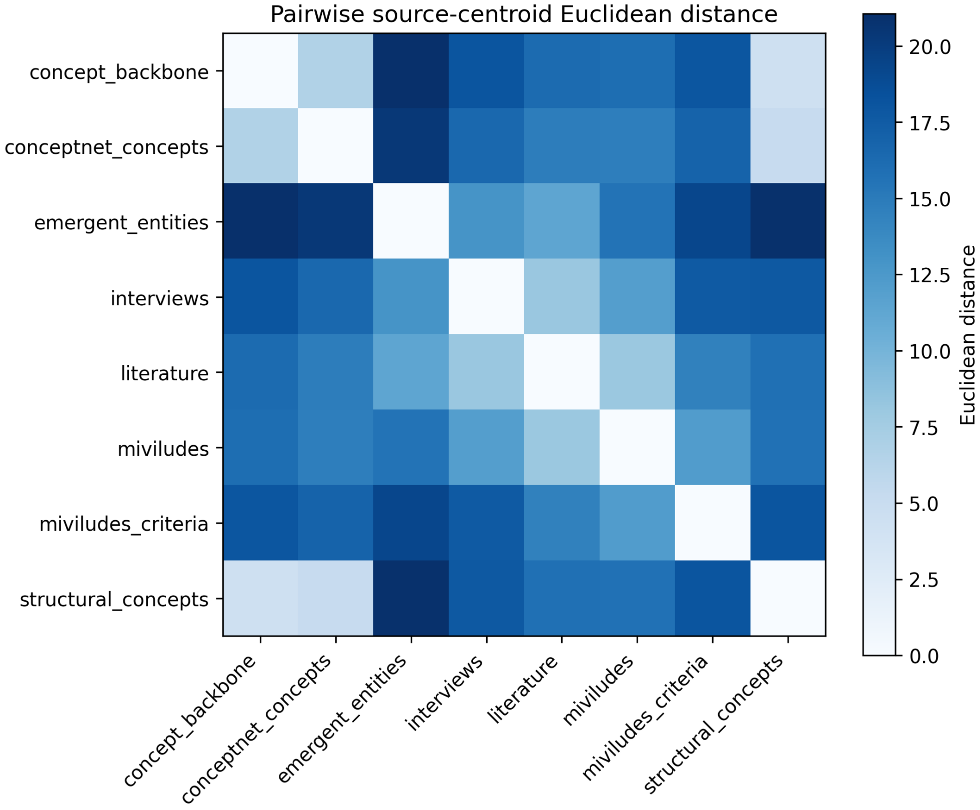

Figure 7: Heatmap of pairwise centroid distances across all eight v2 sources, this run.

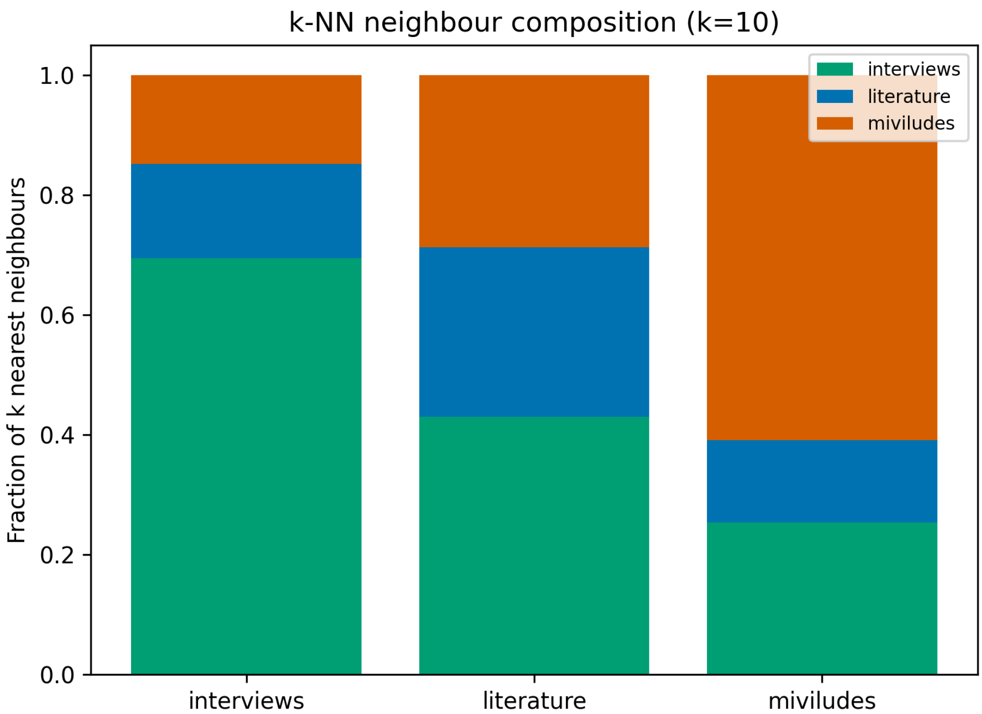

Figure 8: k-NN neighbour composition at $k = 10$, equal-n sampling, this run.

Correlation: Spearman rank correlation between publication year and a point's distance to

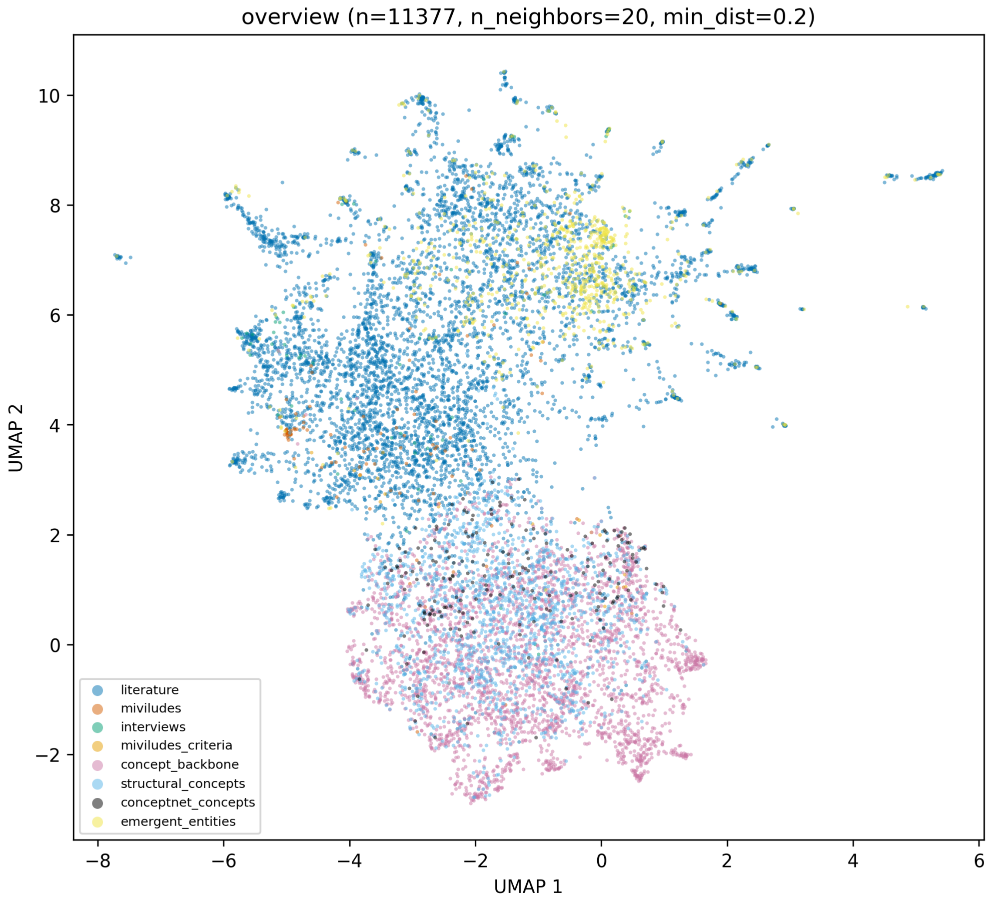

Figure 9: Whole-space UMAP overview, coloured by source dataset, this run (11,377 points).

literature's own centroid: $\rho = -0.064$, $p < 0.0001$ (n=5,741). Statistically significant at this sample size, but a very weak effect – publication year explains almost none of the variation in how "typical" a literature expression is of literature's own centre of gravity.

Period drift: splitting literature into year terciles using the corpus's own distribution (never a fixed calendar grid) gives early (2001–2004, n=2,556), mid (2006–2009, n=1,634), and late (2010–2023, n=1,551) – an uneven split by point count (a handful of large early reference works dominate "early"), flagged as a limitation below, not hidden. Pairwise period-centroid cosine similarities stay high (0.91–0.96) – the three periods' average positions are not dramatically different from each other. Most notably: the single nearest MIVILUDES criterion to each period's own centroid is **the same criterion in all three periods** – *Le changement radical de comportement* ("radical behaviour change") – with distance ranging only 29.07–29.58 across the 2001–2023 span. On this specific measure, literature's closest alignment to MIVILUDES's official framework has not shifted detectably over more than two decades.

11 Honesty notes – what these numbers do not show

- **Corpus imbalance persists**: literature is 97.09% of expression points; equal-n/equal-size corrections remain necessary and are used throughout.
- **The equal-n draw is 64 points**, not v1's 204 – every bootstrapped CI in this document rests on a smaller sample.

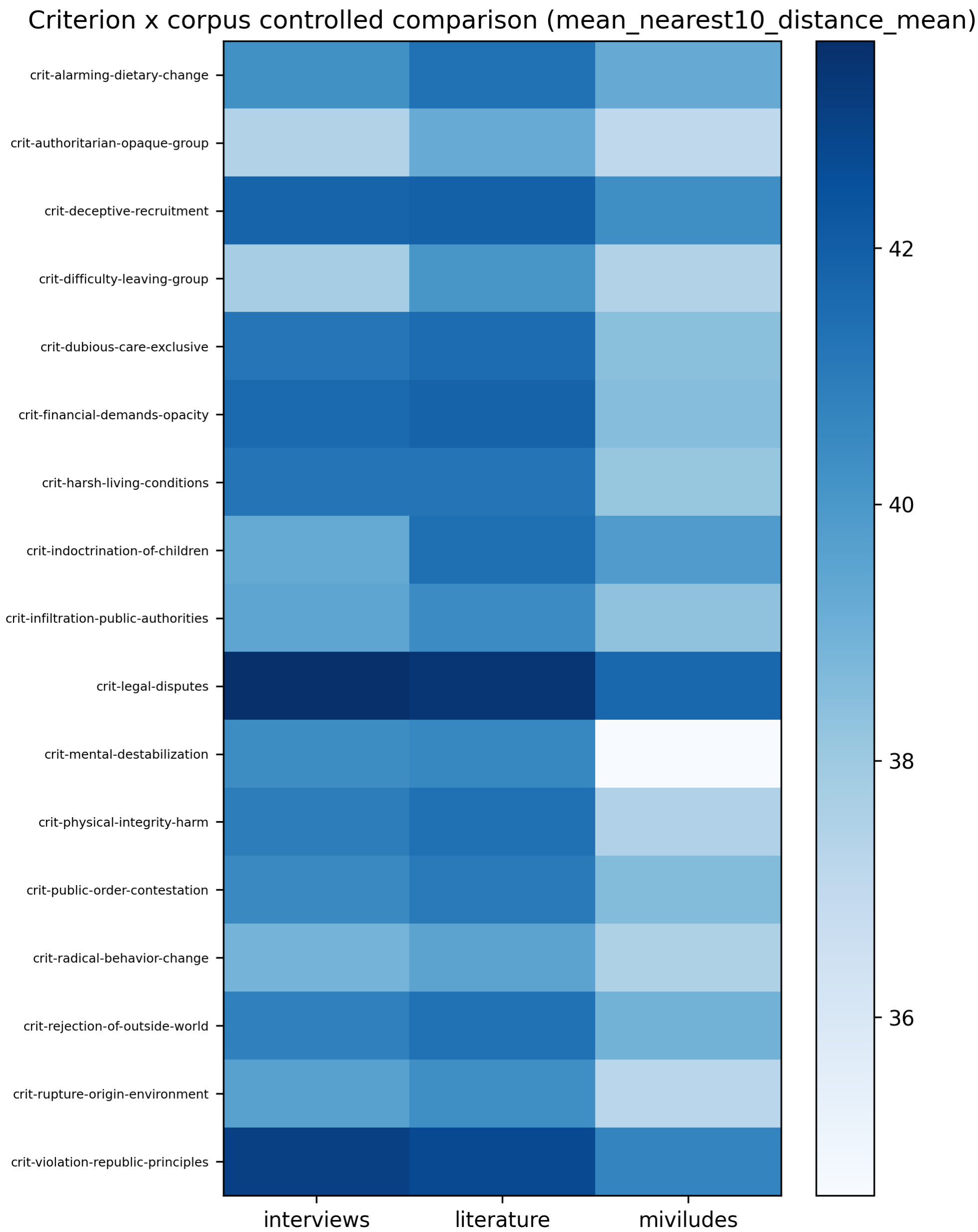

Figure 10: Heatmap of each of the 17 MIVILUDES criteria against the three expression corpora, this run.

- **The entity-layer fix is a substantial improvement, not a finished, hand-verified list.** Casing + per-corpus frequency + a corpus-author stop-list catches most noise; externally-cited classical theorists not among this corpus's own 57 authors are a known, unfixed residual (Section 6).
- **French/English entity variants do not automatically merge.** "Scientology" (English) and "la scientologie" (French) are normalized and counted as separate entities unless the exact same string recurs – so a genuinely shared concept can undercount as `shared_all_3` if each corpus happens to use a different-language form. The 3 `shared_all_3` entities found are a floor, not a ceiling.
- **MIVILUDES is still two documents**, 108 expressions.

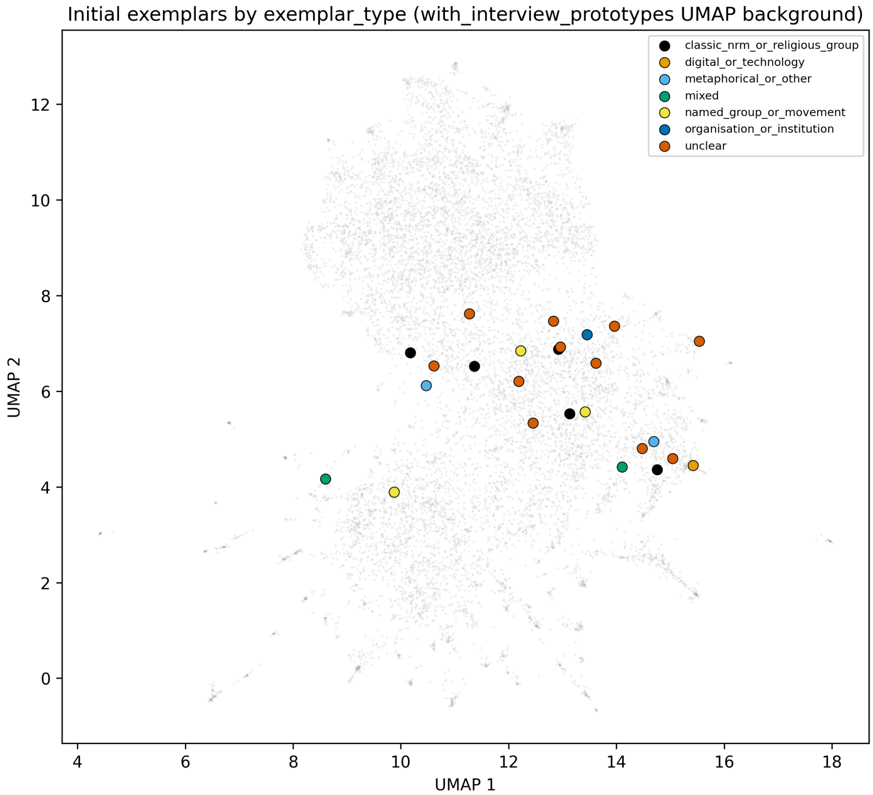

Figure 11: Interview opening responses grouped by exemplar type, this run.

- **The interview sample is a convenience sample**, 63 pooled expressions (64 extracted, 1 manually excluded – "Sept?," below) across 26 people.
- **`response_rank` is still not usable** as a "first thing they thought of" signal.
- **One point ("Sept?") passed every automated check despite carrying no propositional content** – manually excluded this run (Section 5); was 1/11,378, negligible scale either way, but a real, documented case of automated checks not catching everything.
- **Equal-size-controlled comparisons control for pool size only.**
- **2-D pictures are a reading aid, not evidence** – and this run found a concrete case of that mattering: Section 8's criteria-seeded and corpus-centroid-seeded Voronoi diagrams failed to render usefully, purely as an artefact of projecting out-of-sample points into an already-fitted UMAP manifold, with no bearing on the underlying high-dimensional retrieval this document relies on elsewhere.
- **Typicality/borderline scores are purely geometric**, not a claim about cognitive prototypicality or real-world category membership – the same applies to Section 7's cluster/centrality scores.
- **HDBSCAN leaves most entities (68%, 514/753) as "noise"** at the threshold used (min_cluster_size=3) – expected for a heterogeneous set of named things, not evidence those entities are unimportant, but a real limit on how much of the entity layer the cluster-based analyses (Section 7's summary table, the cluster-centroid-seeded Voronoi diagram) actually speak to.
- **One cluster ("california; los angeles; san francisco") looks like a co-occurrence**

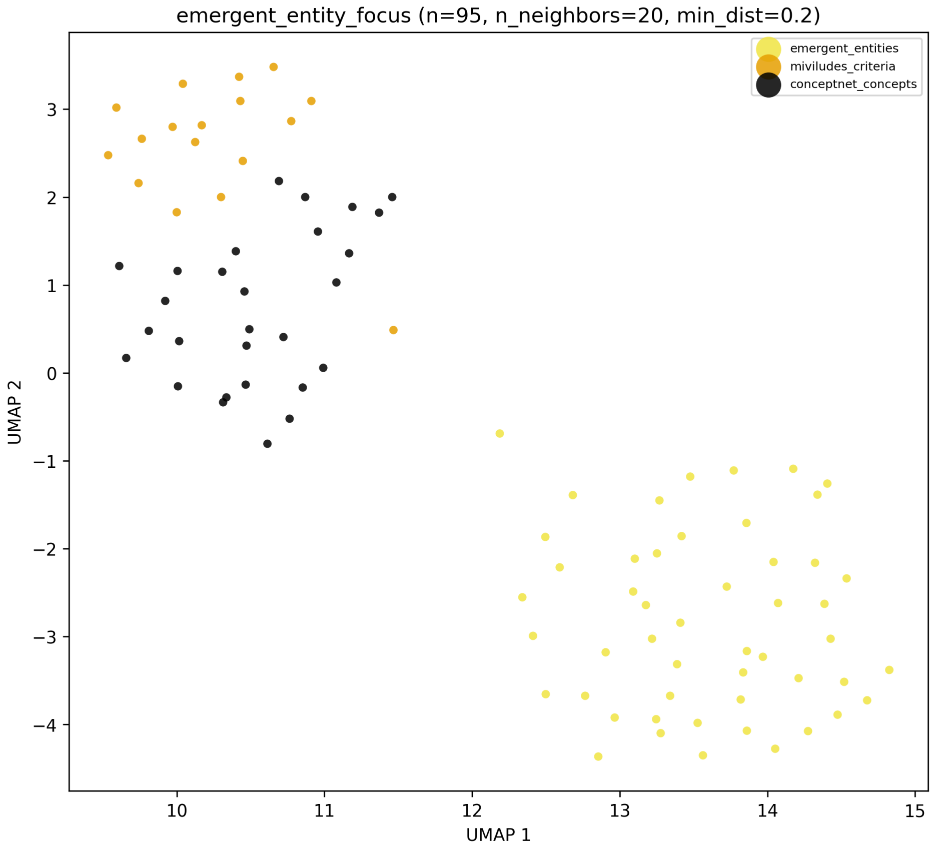

Figure 12: Focused UMAP of the top 50 emergent entities (plus neighbourhood context), coloured by source dataset – the first version of this figure worth showing, now that the entity layer has real content.

artefact, not a meaningful "cult entity" grouping – reported as a likely artefact in Section 7 rather than smoothed over.

- **The publication-year terciles are size-imbalanced** (2,556/1,634/1,551 points) by construction of splitting on the actual year distribution rather than a fixed calendar grid – the "early" period is dominated by a handful of large early reference works, not an equal sample of early-period scholarship.
- **The year/distance correlation is real but very weak** ($\rho = -0.064$): statistically significant at n=5,741, but publication year explains almost none of the actual variation – reported as "a detectable but small effect," not "years matter a lot."

12 Possible interpretations

Read this box before the rest of this section. Everything above this point is descriptive. Everything here is a provisional *reading* of that description, offered because it was explicitly requested as a first pass – not proof. Each is paired with a reason it might be wrong.

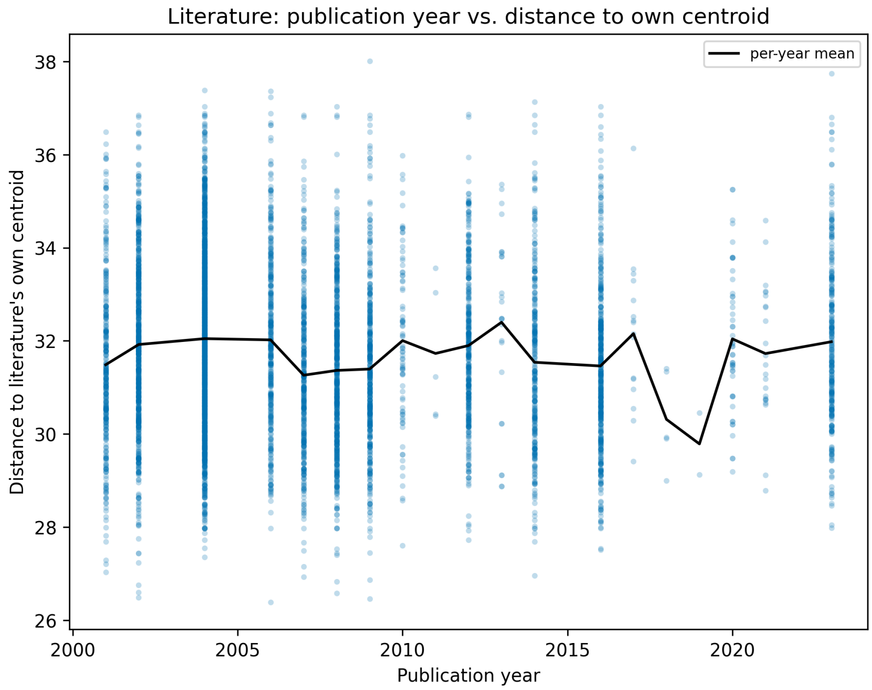

Figure 13: Literature: publication year vs. distance to literature's own centroid, one point per expression, with the per-year mean overlaid.

On what the entity fix itself shows. The gap between 21 and 753 entities – a 36-fold difference from a methodology bug, not a change in the underlying corpus – is itself informative: it demonstrates how sensitive a "count of named things" statistic is to exactly which extraction field it draws from, and by extension how much scrutiny any single such count deserves before being treated as evidence about what a corpus "really" emphasizes. One reading: this argues for treating any single emergent-entity number in this thesis (this run's 753 included) as provisional pending further cross-checks, not as a finished census. **Why this might overstate the point**: the specific bug found here (only extracting entities from already-kept expressions) is a one-off implementation gap tied to this particular two-field extraction schema, not evidence that the corrected 753-entity count itself is still substantially incomplete – the fix targeted a known, specific mechanism, not a general suspicion.

On the 3 shared_all_3 entities. "New Age," Jehovah's Witnesses, and the Order of the Solar Temple being the only entities mentioned by all three corpora – an academic literature review, a French state agency's operational report, and 26 lay interviews – is a narrow but suggestive overlap: two are specific, well-known, historically prominent named groups (one linked to a mass murder-suicide, the Solar Temple), and one is a broad, loosely-bounded movement label. One reading: what these three very differently-motivated sources agree is worth naming at all skews toward either extremely notorious specific cases or extremely broad umbrella categories, with comparatively little shared middle ground. **Why this might be wrong**: with only 3 data points, and given the previous interpbox's point about French/English variants not auto-merging, this could easily be undercounting genuinely shared entities that happen to be phrased differently per corpus (e.g., Scientology is very likely discussed by all three under different exact strings) – 3 is a floor, and the pattern reading above is built on a small, possibly incomplete sample.

On 55 interviews-only entities appearing for the first time. The first v2 pass found zero interview-specific entities; this run finds 55 (Netflix, "AI cult," Blue Öyster Cult, Richard Branson, "Tomato cult," and others), none overlapping literature or MIVILUDES. One reading: lay interviewees' spontaneous associations with "cult" draw on a genuinely different reference set than either academic literature or state institutional framing – pop culture, personal media consumption, loose/metaphorical usage – rather than simply being a smaller, noisier version of the same discourse. **Why this might be wrong**: at min-mentions$\geq$1 for interviews, a single interviewee's one-off remark is enough to create an entity, so some of these 55 may be idiosyncratic to one person rather than any kind of shared "interview discourse" pattern – the count says these entities are *interview-exclusive*, not that they are *common* within interviews.

On generic vocabulary sitting central, specific groups sitting peripheral. Section 7's most-central entity clusters are generic, often multilingual core terms for the topic itself (sect/secte/sekte, "new religious movement"); the most peripheral are highly specific named groups (the Clapham Sect, the Moonies, Jehovah's Witnesses in four languages). One reading: this mirrors exactly the same pattern Section 9's typicality analysis already found for ordinary *expressions* (generic statements sit near a corpus's centroid, named specifics sit at the edge) – suggesting it is a general property of how this embedding space organizes "generic descriptive language" versus "a specific proper noun," not something particular to entities. **Why this might be incomplete**: a generic term's centrality could also partly reflect that MANY different corpus passages independently produced very similar entity strings for it (e.g. "secte"/"sects"/"sekte" are near-synonyms across languages, pulling their shared region toward the centre by sheer repetition), which is a different mechanism from "genuinely more prototypical" – this run's data does not distinguish the two explanations from each other.

On the Voronoi comparison's own methodological lesson. Only the cluster-centroid-seeded strategy worked as a 2-D picture; the externally-imposed criteria and corpus-centroid seeds did not. One reading: this is itself a small, generalizable methodological point worth carrying into other parts of the analysis toolkit – any time this project projects an "external" reference point (a criterion, a centroid, a prototype) into a 2-D layout it was not part of fitting, the same compression risk applies, and should be checked for, not assumed away. Several existing focused-projection figures elsewhere in this toolkit already do exactly this (overlaying criteria/centroids on a UMAP fit to something else) and may be silently affected without this specific failure mode having been checked for them. **Why this might overstate the point**: several of those existing figures fit UMAP on a population that already INCLUDES the criteria/centroids as members (rather than fitting on entities-only and transforming criteria in afterward, as this module does) – which sidesteps the specific out-of-sample mechanism identified here – so this concern needs checking on a per-figure basis, not assumed to generalize.

On the flat publication-year finding. The nearest MIVILUDES criterion to literature's own centroid is identical across all three publication-year periods, and the year/distance correlation is real but weak. One reading: this is mild evidence against a "the academic conversation about cults has fundamentally reoriented since 2001" narrative, at least along this specific geometric measure – the topic's centre of gravity, relative to MIVILUDES's own framework, looks similar in 2001–2004 and 2010–2023. **Why this might be wrong, or at least incomplete**: this is a single aggregate measure (whole-corpus centroid position) that could easily mask real, opposing shifts in different sub-literatures (e.g. a genuine rise in discussion of digital/online radicalization, offset by a genuine decline in discussion of 1970s-80s "classic" NRMs, netting out to a similar overall centroid) – distinguishing "nothing changed" from "several things changed and cancelled out in aggregate" would need a finer-grained, topic-level analysis this run does not attempt.

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13 Using this document to write the Results chapter

Section 6 is unusually long for a "results" report because the correction it documents is itself a methodologically relevant fact – how a bug in a two-field extraction schema silently discarded most named-entity signal, and what three independent, data-checked fixes (casing filter, a filter bug, a corpus-derived stop-list) it took to recover it, is exactly the kind of thing a Methods chapter should be candid about. Sections 7–10 are new this run, built at the researcher's request to see whether a spatial/topological reading of the entity layer, several different ways of partitioning it, and a publication-date check would surface anything the earlier, purely tabular analyses did not – the honest answer is "partially": the entity-cluster and gap analyses produced real, checkable structure, one of three Voronoi strategies worked as a picture (and the other two's failure is itself informative, Section 8), and the publication-date check came back mostly flat. Section 12 remains a first, provisional pass, kept structurally separate from descriptive fact throughout, offered as a starting point for argument, not a finished answer.